

Minimum Moves, Minimal Intelligence Exercise

Executive Summary

The "minimum moves, minimal intelligence" concept can be formalized as a parsimonious problem-solving model: achieve a defined objective with the fewest state-changing actions while using the least cognitive complexity, contextual knowledge, and computational effort necessary to remain reliable.[cite:16][cite:20][cite:24] In cognitive science and decision research, this aligns with cognitive parsimony, the law of less work, and simplicity preferences in human decision-making, which show that people and systems often prefer lower-effort strategies when they are sufficient for the task.[cite:20][cite:24][cite:26]

This model works best in environments that are well-defined, repetitive, high-volume, and constrained, because simple heuristics and low-branching decision rules can produce fast, reliable outcomes without expensive reasoning.[cite:22][cite:31][cite:32] It becomes dangerous when the environment is adversarial, ambiguous, rapidly changing, or safety-critical, because minimizing steps and cognition can suppress adaptation, nuance, and error detection precisely when they matter most.[cite:46][cite:48][cite:66]

The sections below execute the full prompt in three parts: three easy exercises, a business cost-mitigation platform model, and three worst-case nation-state defense scenarios.[cite:20][cite:32][cite:49]

Core Concept Definition

Working Definition of "Minimum Moves"

"Minimum moves" means reaching a specified end state with the smallest possible number of actions that materially change the state of the task or system.[cite:16][cite:22] A move may be physical, informational, procedural, or computational, so long as it advances the problem from its initial state toward its goal state.[cite:16][cite:25] In applied settings, the point is not elegance for its own sake but measurable efficiency: fewer actions generally mean less time, lower cost, fewer handoffs, and fewer opportunities for error.[cite:32][cite:38][cite:41]

Working Definition of "Minimal Intelligence"

"Minimal intelligence" means solving a problem with the lowest sufficient level of reasoning, memory, search, abstraction, and contextual interpretation required for success.[cite:16][cite:20][cite:24] In practice, this means favoring simple heuristics, rules of thumb, finite decision trees, templates, and bounded procedures before using more complex reasoning or advanced models.[cite:22][cite:31][cite:42] This follows the parsimony principle: avoid unnecessary complexity when a simpler method explains, decides, or acts well enough.[cite:17][cite:23][cite:24]

Combined Principle

Together, the model asks two questions: what is the fewest actions needed, and what is the least intelligence needed to choose and execute those actions reliably.[cite:20][cite:24][cite:31] The model therefore optimizes for "good enough with minimal overhead," not for maximum sophistication.[cite:31][cite:37][cite:74] It is strongest when tasks are standardized and weakest when hidden variables, deception, or cascading consequences punish oversimplification.[cite:36][cite:46][cite:49]

Exercises

Exercise 1: Shortest Grid Path

Objective: reach a goal square on a small grid in the fewest possible moves using only local movement rules.[cite:22][cite:62]

Setup

- Draw a 4x4 grid on paper.[cite:62]
- Mark one square as **Start** and one as **Goal**. [cite:62]
- Place 2 blocked squares as obstacles.[cite:62]
- Difficulty: **very easy**. [cite:62]

Allowed Moves

- Move exactly one square up, down, left, or right per turn.[cite:62]
- No diagonal moves.[cite:62]
- The participant may only look one move ahead and may not sketch alternate plans.[cite:62][cite:74]

Success Condition

Success is reaching the goal square.[cite:22] The minimum-moves solution is the shortest available route measured in grid steps.[cite:22][cite:62] The minimum-intelligence condition is satisfied if the solver uses a simple rule such as "always reduce Manhattan distance unless blocked; if blocked, go around the obstacle in the first available direction." [cite:22][cite:74]

Example Walkthrough

Initial state: Start at lower-left corner, Goal at upper-right corner, with blocks at coordinates (2,2) and (2,3). A valid minimal path is Right, Right, Right, Up, Up, Up for six moves if the obstacle layout permits that corridor; if the direct path is blocked, the solver uses the same greedy rule until the obstacle is encountered and then takes the shortest detour around it.[cite:22][cite:62] This exercise remains easy because the rule set is tiny, the state space is visible, and the participant does not need deep search or memory.[cite:16][cite:22]

Assessment Notes

A tester can score total moves, excess moves above the shortest path, hesitation time, and number of unnecessary reversals.[cite:20][cite:62] The boundary between minimal and higher intelligence appears when the participant begins simulating many alternate futures instead of applying a local

heuristic.[cite:20][cite:24]

Exercise 2: Two-Bucket Sorting

Objective: place a short list of numbers into two bins, "small" and "large," with the fewest checks and no calculation beyond a single threshold comparison.[cite:22][cite:62]

Setup

- Write eight numbers on cards, for example: 1, 3, 4, 5, 7, 8, 9, 10.[cite:62]
- Define the rule: numbers 1–5 go in **Small**, and 6–10 go in **Large**.[cite:62]
- Difficulty: **very easy**.[cite:62]

Allowed Moves

- Pick up one card.[cite:62]
- Compare it to the threshold rule.[cite:62]
- Place it into one bucket.[cite:62]
- No rechecking after placement unless the tester allows one correction move.[cite:62][cite:74]

Success Condition

Success is correct classification of all items.[cite:22] The minimum-moves path is one check and one placement per card, with no resorting.[cite:62] The minimum-intelligence condition is met because the participant only needs a single binary rule: "If the number is 5 or less, place in Small; otherwise place in Large." [cite:62][cite:74]

Example Walkthrough

Given the sequence 7, 3, 10, 1, 5, 8, 4, 9, the participant performs eight comparisons and eight placements.[cite:62] Card 7 goes to Large; card 3 goes to Small; card 10 goes to Large; card 1 goes to Small; card 5 goes to Small; card 8 goes to Large; card 4 goes to Small; card 9 goes to Large.[cite:62] The procedure is minimal because every item requires exactly one decision and one placement, and no deeper reasoning improves the result.[cite:20][cite:74]

Assessment Notes

A tester can measure cards processed per minute, misclassifications, and whether the participant uses forbidden additional strategies such as grouping, recounting, or memorizing patterns in advance.[cite:20][cite:62] This exercise shows how many routine tasks can be handled by threshold logic rather than true analysis.[cite:31][cite:42]

Exercise 3: One Question Locator

Objective: identify which of two hidden locations contains a target by asking the minimum number of yes/no questions.[cite:22][cite:62]

Setup

- A tester hides an object in either Box A or Box B.[cite:62]

- The participant is allowed one yes/no question before making a choice.[cite:62]
- Difficulty: ****novice****. [cite:62]

Allowed Moves

- Ask one yes/no question.[cite:62]
- Choose Box A or Box B.[cite:62]
- Open one box.[cite:62]

Success Condition

Success is selecting the correct box.[cite:22] The minimum-moves solution is one discriminating question plus one selection.[cite:62] The minimum-intelligence condition is satisfied if the participant asks the most informative possible question, such as "Is it in Box A?" rather than introducing indirect wording or multi-step logic.[cite:22][cite:62]

Example Walkthrough

The participant asks, "Is the object in Box A?" If the answer is yes, they choose Box A; if no, they choose Box B.[cite:62] This is minimal because a single binary question fully resolves a two-state uncertainty.[cite:22][cite:74] It illustrates a core lesson of the model: when the decision space is already compressed, sophisticated reasoning adds no value.[cite:24][cite:31]

Assessment Notes

The tester records whether the question was maximally informative, whether the participant wasted the question, and whether they changed strategy unnecessarily.[cite:20][cite:62] This exercise is useful for training interview prompts, diagnostic trees, and automated triage logic.[cite:31][cite:42]

Why These Three Are the Easiest

These exercises are the easiest because they are fully bounded, visible, and reducible to one dominant heuristic each: shortest path, threshold split, and binary elimination.[cite:22][cite:62][cite:74] They require almost no domain knowledge, minimal working memory, and limited branching, which makes them suitable for human training, simple AI benchmarking, or low-cost automation design.[cite:16][cite:20][cite:31]

Business Cost-Mitigation Model

Overview

A business can use this concept as a primary cost-mitigation platform by treating every application workflow as a decision path that should be resolved in the fewest valid steps by the least-expensive adequate intelligence layer.[cite:31][cite:32][cite:36] This closely matches lean process improvement, where organizations map value streams, remove non-value-added steps, and continuously reduce effort, time, cost, and mistakes.[cite:32][cite:38][cite:41] In AI and automation terms, it also resembles Minimum Viable Intelligence: using only as much intelligence as needed to solve the business problem at acceptable cost and risk.[cite:31][cite:37]

The Platform Logic

The platform should be built on four tiers.

1. Input Normalization Layer

All incoming work items should be converted into a narrow, structured format: intent, urgency, category, confidence, and required outcome.[cite:33][cite:42] This matters because simple decision systems perform best when inputs are standardized and ambiguity is reduced before the actual routing logic runs.[cite:31][cite:36]

2. Minimal-Intelligence Decision Layer

A rules engine or decision engine should handle the majority of common cases using decision tables, branching rules, eligibility checks, confidence thresholds, and standard response templates.[cite:36][cite:39][cite:42] This layer is the core cost-mitigation mechanism, because it replaces expensive generalized reasoning with cheap, transparent, repeatable logic for high-volume tasks.[cite:31][cite:36][cite:39]

3. Atomic Action Layer

Every approved output should be expressed as a small set of atomic actions, such as send response, update record, request missing document, route to queue, or trigger payment review.[cite:32][cite:38] Atomic actions make moves countable, auditable, and optimizable, which is essential if the business wants to reduce average move count per transaction over time.[cite:36][cite:39]

4. Escalation Layer

Anything outside rule coverage, below confidence threshold, above risk tolerance, or involving compliance ambiguity should escalate to a higher-intelligence tier such as a human specialist or advanced AI model.[cite:31][cite:36][cite:42] The model fails when organizations try to force all cases through the cheap layer, so escalation is not a defect but a required safety valve.[cite:31][cite:77][cite:85]

Operating Model for a Mid-Size Business

A mid-size firm can implement this as a "cost-aware orchestration platform" that sits between incoming demand and final action.[cite:31][cite:36][cite:42] The core KPI is not only speed but ****cost per resolved case****, broken down by move count, labor minutes, rework rate, and escalation frequency.[cite:32][cite:38][cite:40] Over time, the business should push more work into the low-cost tier only when that tier consistently meets quality, compliance, and customer outcome thresholds.[cite:31][cite:40]

Example: Customer Support Triage

Customer support is a strong fit because a large share of tickets are repetitive, low-context, and resolvable through simple logic.[cite:31][cite:34][cite:39] A support request enters the system, is classified into a narrow intent class such as password reset, refund status, shipping issue, or billing address change, and is then routed through a rules engine that chooses the smallest complete response path.[cite:33][cite:39][cite:42]

A password reset case is illustrative. The system confirms identity, checks whether self-service is available, sends the reset workflow, and closes the case if the user completes it; if identity cannot be verified or the account is flagged, the case escalates automatically.[cite:36][cite:39][cite:42] That path minimizes both moves and intelligence because it uses a small number of deterministic checks instead of a broad conversational AI model or a live human agent for every case.[cite:31][cite:37]

Example: Back-Office Document Processing

Invoice review, claims handling, and form validation also fit this framework well.[cite:34][cite:40] The system first checks whether required fields are present, whether values fall within expected ranges, and whether policy rules are satisfied; only exceptions are sent for advanced review.[cite:32][cite:36][cite:41] This lowers labor demand because staff spend less time on straightforward approvals and more time on genuinely ambiguous or high-risk work.[cite:34][cite:40]

Cost-Mitigation Mechanisms

This model cuts cost through five linked mechanisms:

- **Lower labor intensity:** repetitive cases are resolved without human review whenever simple logic is enough.[cite:31][cite:34][cite:40]
- **Fewer handoffs:** lean mapping shows that non-value-added approvals and transfers create delay and cost.[cite:32][cite:38][cite:44]
- **Lower training burden:** staff working on escalated cases can focus on exceptions rather than memorizing routine scripts.[cite:31][cite:36]
- **Lower rework:** rule-driven workflows standardize outputs and reduce variance.[cite:36][cite:39]
- **Faster change management:** centralizing policy logic in a rules engine avoids modifying multiple application code paths when rules change.[cite:36][cite:42]

Risks and Trade-Offs

The same simplifications that cut cost also create brittle points.[cite:31][cite:77][cite:85] If input normalization is poor, if decision rules are obsolete, or if edge cases are incorrectly forced through the low-intelligence tier, the business may save money on routine cases while creating hidden losses through compliance failures, customer dissatisfaction, or exception backlogs.[cite:36][cite:39][cite:77] The model therefore works best when paired with governance, audit logging, rollback ability, and explicit human override paths.[cite:39][cite:77]

Implementation Roadmap

1. **Map the current workflow** and identify high-volume, low-variance tasks.[cite:32][cite:38][cite:41]
2. **Measure current moves** per task, including approvals, transfers, wait states, and rework loops.[cite:38][cite:44]
3. **Define a minimum-move target path** for routine cases.[cite:31][cite:32]
4. **Implement a rules or decision engine** separate from application code.[cite:36][cite:42]
5. **Create confidence thresholds and escalation rules** for ambiguity, risk, or policy conflicts.[cite:31][cite:77]

6. ****Track move count, resolution cost, exception rate, and customer outcomes**** to avoid optimizing only for cheapness.[cite:38][cite:40][cite:77]

Nation-State Defense Scenarios

Scenario 1: Strategic Warning Collapse and Launch-on-Warning Pressure

In the worst case, a nuclear-armed state experiences degraded warning systems, cyber interference, and fragmented communications while receiving indications of an inbound strategic attack.[cite:46][cite:63][cite:66] Under launch-on-warning logic, national leaders may have only minutes to assess whether the warning is real before deciding whether to retaliate.[cite:63][cite:66][cite:72] Multiple policy analyses argue that this compressed decision time is inherently dangerous because it increases the probability of mistaken, ill-considered, or unauthorized launch.[cite:63][cite:66][cite:75]

Why Minimal Intelligence Is Forced

In such a crisis, the state may be unable to run deep verification, broad scenario analysis, or complex deliberation because command-and-control systems are partially degraded and the timeline is measured in minutes.[cite:46][cite:66][cite:72] The nation is therefore pushed toward a minimal-intelligence model: fixed thresholds, predelegated options, and short decision trees such as verify-warning, assess survivability, and execute preplanned response package.[cite:63][cite:72]

Moves and Time Constraint

The relevant moves may be limited to confirm alert, consult a tiny decision group, issue a coded order, or stand down.[cite:63][cite:72] The time horizon can be so short that leaders cannot gather new evidence beyond what the damaged warning network already provides.[cite:66][cite:69]

Failure Modes

This is catastrophic because the model minimizes deliberation precisely when deception, technical malfunction, or spoofed data are possible.[cite:46][cite:66] Analysts note that short warning and decision time, combined with cyber threats to warning and command systems, magnifies the risk of nuclear use based on false or ambiguous signals.[cite:66][cite:69][cite:75] In this scenario, "minimum moves" can become "minimum safeguards," producing civilization-scale consequences from a low-information trigger.[cite:63][cite:75]

Mitigation and Residual Risk

The best mitigation is structural, not procedural: increase warning and decision time, reduce prompt-launch dependence, and harden crisis communications so leaders are not trapped in an ultra-short response loop.[cite:46][cite:66][cite:72] Even then, residual risk remains if adversaries deliberately target communications and early-warning reliability during crises.[cite:46][cite:66]

Scenario 2: Theater War Under Denied, Degraded, and Disrupted Operations

A nation engaged in high-end conventional war may lose GPS, satellite communications, ISR feeds, and portions of its networked command structure because a near-peer adversary is jamming, spoofing, or disabling space-enabled capabilities.[cite:48][cite:49][cite:51] U.S. military sources explicitly warn that current forces must learn to operate when common domains become unusable or inaccessible,

and Army training materials describe denied, degraded, and disrupted space environments as realistic features of near-peer conflict.[cite:48][cite:49]

Why Minimal Intelligence Is Forced

When the network collapses, commanders cannot rely on rich data fusion, centralized optimization, or full-spectrum situational awareness.[cite:48][cite:49][cite:52] They revert to minimal-intelligence tools: standard operating procedures, battle drills, mission-type orders, commander's intent, and predetermined branch plans.[cite:49][cite:52][cite:58]

Moves and Time Constraint

Moves may include reposition unit, retransmit via line-of-sight systems, execute preplanned fires, or disperse to fallback positions.[cite:49][cite:54] Time constraints range from seconds for local survivability to hours for maneuver and resupply, but decisions still occur with far less information than peacetime doctrine assumes.[cite:48][cite:49]

Failure Modes

This model can preserve action, but it also creates dangerous rigidity.[cite:48][cite:52][cite:58] If the adversary understands the defender's playbook, fixed responses become predictable, units may misread local conditions, and decentralized actors may act on stale intent because they cannot confirm changes in the wider operational picture.[cite:49][cite:52] The result can be fratricide, mistimed fires, exposed logistics nodes, or wasted combat power committed according to yesterday's assumptions.[cite:48][cite:49][cite:58]

Mitigation and Residual Risk

Mitigation requires prewar training for degraded operations, distributed decision competence, resilient local communications, and mission command cultures that tolerate initiative without requiring constant central supervision.[cite:48][cite:49][cite:52] Residual risk remains high because a capable adversary can keep attacking the very systems that would restore adaptive intelligence.[cite:49][cite:51]

Scenario 3: Homeland Critical Infrastructure Crisis Managed by Simple Automated Rules

A nation may face a coordinated cyberattack or EMP-like disruption that simultaneously degrades electricity, telecommunications, water systems, transportation, and emergency response coordination.[cite:47][cite:53][cite:76] Intelligence and security reporting on infrastructure attacks emphasizes that disruptions to critical services can escalate rapidly into public safety emergencies and cascading system failures.[cite:47][cite:50][cite:56] EMP assessments further warn that large-scale disruption to power and communications could halt military and civilian activity for extended periods and overwhelm existing management capacity.[cite:76][cite:81][cite:87]

Why Minimal Intelligence Is Forced

In a large cascading failure, central authorities may no longer have functioning data pipelines or decision bandwidth for dynamic optimization.[cite:76][cite:81] They are pushed toward simple local rules: isolate compromised networks, shed load, prioritize hospitals and water, ration fuel, and restore services in predefined order.[cite:50][cite:76][cite:77]

Moves and Time Constraint

Moves are operationally small but nationally consequential: disconnect segment, switch to manual mode, activate backup generation, restrict demand, and dispatch emergency crews.[cite:50][cite:76] Time pressure is immediate because attacker breakout times in modern cyber incidents can be under 30 minutes and service failures can propagate across interdependent infrastructure far faster than senior leadership can deliberate.[cite:79][cite:50]

Failure Modes

This is a worst-case reliance on the model because simple rules can contain damage only when the environment behaves within anticipated bounds.[cite:50][cite:77][cite:85] If dependencies are poorly mapped, operators may isolate the wrong systems, starve critical services, or fail to notice that the attack is manipulating the automated rule layer itself.[cite:56][cite:77][cite:79] In the extreme case, a state trying to preserve services through simple heuristics may accidentally deepen a blackout, disable medical delivery chains, or create panic through inconsistent regional actions.[cite:47][cite:53][cite:81]

Mitigation and Residual Risk

Mitigation includes predefined priority ladders, manual override capacity, resilient fallback communications, AI and automation security controls, and regular cross-sector exercises based on cascading infrastructure failure.[cite:50][cite:77][cite:87] Residual risk remains substantial because the more interconnected the infrastructure, the harder it is for low-intelligence local rules to see second-order and third-order effects in real time.[cite:47][cite:50][cite:85]

Where the Model Works Well

The model is strongest when tasks are repetitive, rules are stable, outcomes are binary or low-dimensional, and the cost of delay exceeds the value of nuance.[cite:31][cite:32][cite:74] That is why it performs well in beginner exercises, transactional business workflows, and some tightly bounded operational drills.[cite:20][cite:31][cite:42] In those settings, minimizing moves and intelligence reduces waste without meaningfully sacrificing performance.[cite:32][cite:38]

Where the Model Breaks Down

The model breaks down when the problem space is adversarial, deceptive, nonlinear, politically sensitive, or safety-critical.[cite:46][cite:48][cite:66] In those environments, extra reasoning is not waste; it is a protective function that detects false signals, hidden dependencies, and emergent consequences.[cite:46][cite:50][cite:66] The greatest danger is confusing cost-efficiency with strategic adequacy.[cite:31][cite:77][cite:85]

Final Assessment

The "minimum moves, minimal intelligence" approach is best understood as a disciplined parsimony framework, not as a universal doctrine.[cite:17][cite:20][cite:24] It is useful when the designer can clearly separate routine cases from exceptional ones and when escalation paths are robust.[cite:31][cite:36][cite:42] It becomes dangerous when leaders, businesses, or states use it not as a first-pass efficiency layer but as a substitute for judgment in complex, contested, or catastrophic conditions.[cite:46][cite:66][cite:77]